

Roll No.

THV-401

B. TECH. (ME) (FOURTH SEMESTER)
END SEMESTER EXAMINATION, June/July, 2022

**UNIVERSAL HUMAN VALUES : UNDERSTANDING
HARMONY**

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Explain the basic guidelines for Value Education. What is the need for these guidelines ? (CO1)
- (b) Explain the process of self-exploration. What is the expected result of self-exploration ? (CO1)
- (c) Why is right understanding required in relationship for mutual happiness ? Illustrate with the help of *two* examples from your life. (CO1)
2. (a) Distinguish between natural acceptance and acceptance with few examples. (CO2)

P. T. O.

(2)

- (b) Distinguish between 'animal consciousness' and 'human consciousness'. (CO2)
- (c) Explain what is meant by right utilization of body. (CO2)
- 3. (a) What is the building block for harmony in the society ? Explain with examples. (CO3)
- (b) What is the meaning of universal human order ? What is its scope ? How is the family order related to universal human order ? (CO3)
- (c) Write a short note on the three aspects of preservation. (CO3)
- 4. (a) The four orders in nature have been classified on the basis of their distinct activity, innateness, natural characteristics and inheritance. Explain with *one* example for each one. (CO4)
- (b) Units are in co-existence in space. Where there is a unit, apace is also there. Explain these two statements. (CO4)
- (c) What is the role of material order and bio-order in the fulfillment of human needs ? How does understanding of existence help in right identification of the above ? (CO4)
- 5. (a) What do you understand by the term 'profession' and 'professional ethics' ? (CO5)
- (b) What do you understand by term 'holistic development' and 'universal human order' ? (CO5)
- (c) What do you understand by the term 'holistic technology' and 'holistic management system' ? (CO5)

Roll No.

TME-403

**B. TECH. (ME) (FOURTH SEMESTER)
END SEMESTER EXAMINATION, June/July, 2022**

MANUFACTURING PROCESS-II

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) In an orthogonal cutting operation, the tool has a rake angle = 15° . The chip thickness before the cut = 0.30 mm and the cut yields a deformed chip thickness = 0.65 mm. Calculate (i) the shear plane angle, and (ii) the shear strain for the operation. (CO1)
- (b) A cylindrical work part 200 mm in diameter and 700 mm long is to be turned in an engine lathe. Cutting speed = 2.30 m/s, feed = 0.32 mm/rev, and depth of cut = 1.80 mm. Determine (i) cutting time, and (ii) metal removal rate. (CO1)

P. T. O.

- (c) In a production turning operation, the foreman has decreed that a single pass must be completed on the cylindrical work piece in 5.0 min. The piece is 400 mm long and 150 mm in diameter. Using a feed = 0.30 mm/rev and a depth of cut = 4.0 mm, what cutting speed must be used to meet this machining time requirement. (CO1)
2. (a) A gun drilling operation is used to drill a 7/16 in diameter hole to a certain depth. It takes 4.5 min to perform the drilling operation using high-pressure fluid delivery of coolant to the drill point. The cutting conditions are $N = 300$ rev/min at a feed = 0.001 in/rev. To improve the surface finish in the hole, it has been decided to increase the speed by 20% and decrease the feed by 25%. How long will it take to perform the operation at the new cutting conditions ? (CO2)
- (b) In a single pass drilling operation, a through hole of 15 mm diameter is to be drilled in a steel plate of 50 mm thickness. Drill spindle speed is 500 rpm, feed is 0.2 mm/rev and drill point angle is 118° . Assuming 2 mm clearance at approach and exit, what is the total drill time (in seconds) ? (CO2)/(CO6)
- (c) The time taken to drill a hole through a 25 mm thick plate with their drill rotating at 300 rpm and moving at a feed rate of 0.25mm/rev will be what ? (CO2)
3. (a) Describe the type of milling cutters (any *three*) with relevant applications. (CO3)
- (b) Draw a neat sketch and explain the working of milling process. (CO3)
- (c) Write the nomenclature of grinding wheel. (CO3)/(CO6)

(3)

4. (a) Two pipes of inner diameter 200 mm and outer diameter 220 mm each are joined by flash welding using 30 V power supply. At the interface, 2 mm of material melts from each pipe which has a resistance of 35Ω . If the unit melt energy is 60 J/mm^3 , then time required for welding (in seconds) is (CO4)
- (b) Spot welding of two 1 mm thick sheets of steel (density = 8000 kg/m^3) is carried out successfully by passing a certain amount of current for 0.1 second through the electrodes. The resultant weld nugget formed is 10 mm in diameter and 3 mm thick. If the latent heat of fusion of steel is 1500 kJ/kg and the effective resistance in the welding operation in 100 the current passing through the electrodes is approximately (CO4)
- (c) Explain in details the process of submerged arc welding (SAW) with neat sketch. (CO4/CO6)
5. (a) Differentiate between unconventional machining and conventional machining, state at least *five*. (CO1/CO5)
- (b) Explain the working principle of Ultra Sonic Machining (USM) along with suitable diagram. (CO5)
- (c) Describe in details the working principle of electric discharge machining (EDM) with a neat sketch. (CO5)

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TME-404

B. TECH. (ME) (FOURTH SEMESTER)
END SEMESTER EXAMINATION, June/July, 2022
MEASUREMENT AND INSTRUMENTATION

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What do you mean by subdivision of standards ? Distinguish between primary, secondary and working standards. (CO1)
- (b) Write short notes on the following : (CO5)
 - (i) Bridgman gauge
 - (ii) Proving rings
- (c) Four end bars P, Q, R, S are to be calibrated using a calibrated length bar of 400 mm whose actual length is 399.9998 mm. The bar Q is longer than bar P by 0.0004 mm, bar R is longer than bar P by 0.0003 mm, while bar S is shorter than P by 0.0001 mm. The four gauges together have a combination length of 400.0002 mm. Determine the corrected (actual) length of each end bar. (CO1)

P. T. O.

2. (a) With the help of neat sketch, explain the construction, working and applications of Solex Comparator. (CO3)
- (b) Derive the expression for gauge factor of the strain gauges in terms of Poisson's ratio. (CO6)
- (c) Design a plug and ring gauge to control the production of a 90 mm shaft and hole part of H8e9. Data given :
- $$i = 0.45 D^{1/3} + 0.001 D$$
- the upper deviation for "e" shaft = $-11 D^{0.41}$. The value for standard tolerance grade IT8 = $25i$ and IT9 = $40i$. 90 mm lies in the diameter step of 80 mm and 100 mm. (CO2)
3. (a) Explain the following methods for measuring effective diameter with the help of derivation : (CO3)
- (i) Two wire method
 - (ii) Three wire method
- (b) Briefly describe the gear tooth Vernier Caliper method with the help of derivation to determine the chord length and height of gear tooth. (CO3)
- (c) Explain the following terms : (CO4)
- (i) Sensitivity
 - (ii) Hysteresis
 - (iii) Threshold
 - (iv) Pitch of gear
 - (v) Calibration
4. (a) Explain the principle of interference. Explain, how interference fringes are formed when optical flat is placed on a surface to be tested ? (CO4)

(3)

- (b) (i) Write a short note on Cathode Ray Oscilloscope (CRO).
 - (ii) Differentiate between mechanical and electrical transducers. (CO6)
 - (c) (i) Explain Optical pyrometer.
 - (ii) Explain the laws of thermocouple. (CO6)
5. (a) What do you mean by Analytical Balance ? Derive the expression for its sensitivity. (CO5)
- (b) Explain the construction and working of the following : (CO5)
- (i) McLeod gauge
 - (ii) Prony brake dynamometer
- (c) Explain the terminology of screw thread with the help of a neat sketch. (CO3)

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TME-405

**B. TECH. (ME) (FOURTH SEMESTER)
END SEMESTER EXAMINATION, June/July, 2022**

KINEMATICS OF MACHINES

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

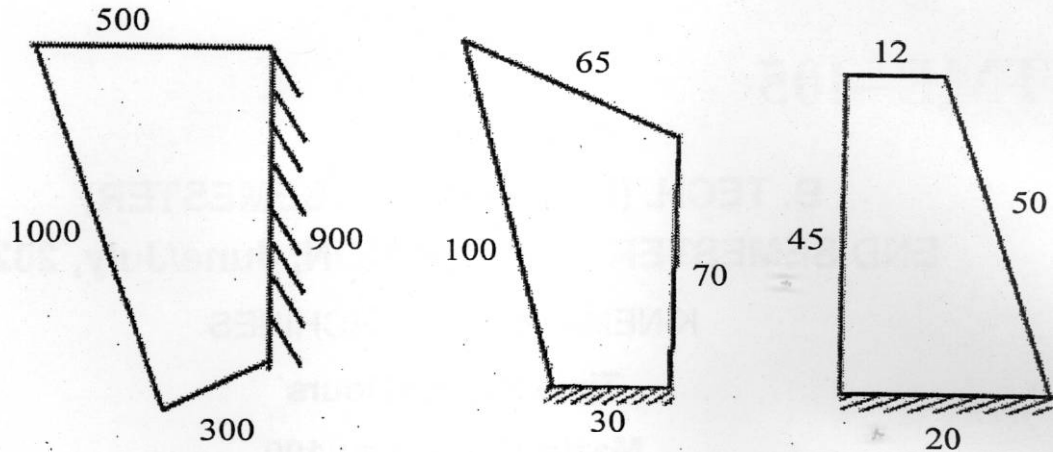
(iv) Each sub-question carries 10 marks.

1. (a) What are the different types of kinematic pairs ? (CO1)
- (b) Write difference between Machine, Mechanism and Structure. Also calculate the degree of freedom of the mechanism with six quaternary links, seven binary joints and one higher pair. Also identify the nature of kinematic chain. (CO1)
- (c) Define Inversions of Single Slider Kinematic Chain with proper diagrams (any *two*). (CO1)
2. (a) A circle has OR as its diameter and a point Q lies on its circumference. Another point P lies on the line OQ produced. If OQ turns about O as centre and the product $OQ \times OP$ remains constant, show that the point P moves along a straight line perpendicular to the diameter OR. (CO2)

P. T. O.

(b) Determine the type of mechanism shown in figure with valid reason :

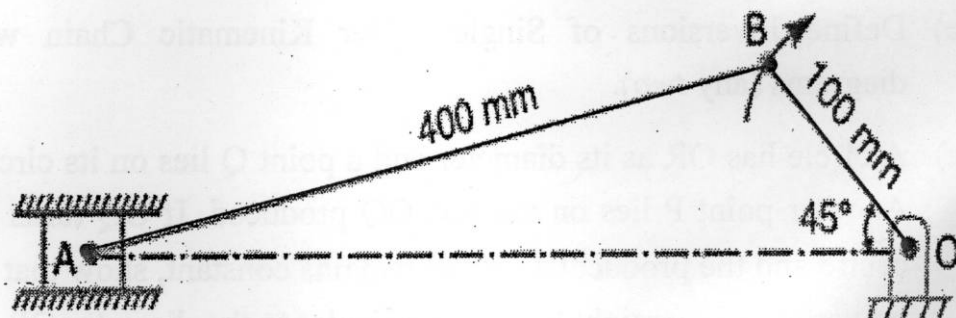
(CO2)



(c) Explain the working of Geneva mechanism and Robert's straight line mechanism. (CO2)

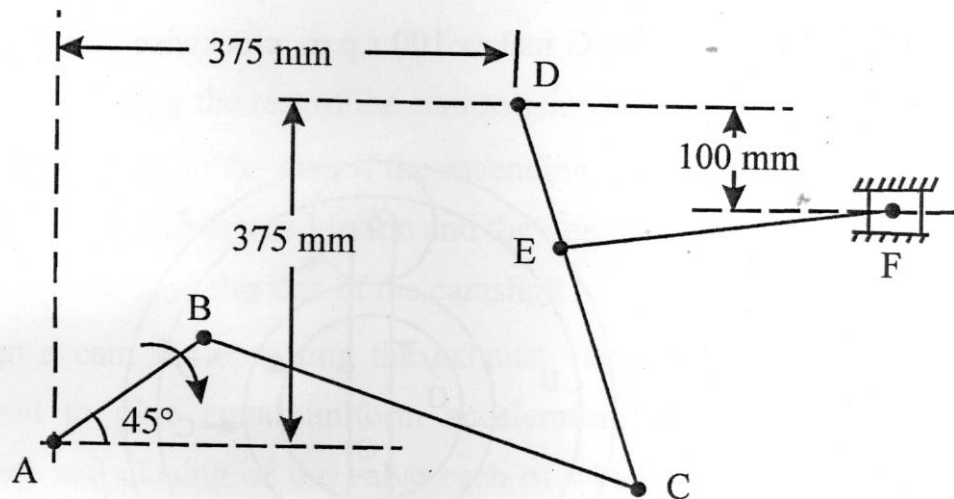
3. (a) Locate all the instantaneous centres of the slider crank mechanism as shown in figure below. The lengths of crank OB and connecting rod AB are 100 mm and 400 mm respectively. If the crank rotates clockwise with an angular velocity of 10 rad/s, find : (CO4)

- (i) Velocity of the slider A, and
- (ii) Angular velocity of the connecting rod AB.



- (b) Define Coriolis acceleration. Derive an expression for magnitude of Coriolis acceleration. (CO3)
- (c) The mechanism, as shown in Fig. below, has the dimensions of various links as follows : (CO3)

$AB = DE = 150 \text{ mm}$; $BC = CD = 450 \text{ mm}$; $EF = 375 \text{ mm}$.

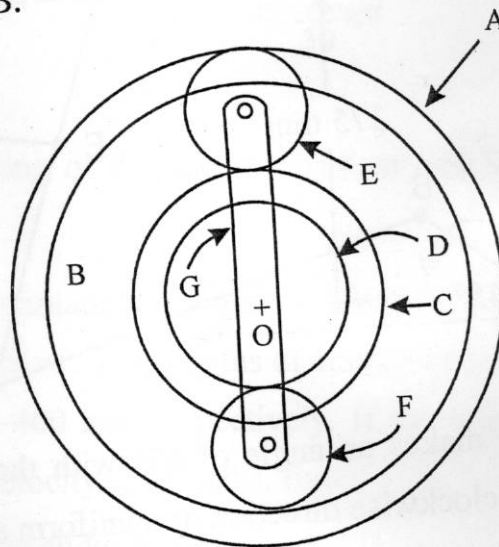


The crank AB makes an angle of 45° with the horizontal and rotates about A in the clockwise direction at a uniform speed of 120 r.p.m. The lever DC oscillates about the fixed point D, which is connected to AB by the coupler BC.

The block F moves in the horizontal guides, being driven by the link EF. Determine : (i) velocity of the block F, (ii) angular velocity of DC, and (iii) rubbing speed at the pin C which is 50 mm in diameter.

4. (a) Explain any *ten* gear tooth terminologies with suitable diagram. (CO5)
- (b) Define and explain the condition for constant velocity ratio during gear meshing. (CO5)

- (c) In an epicyclic gear train, the internal wheels A and B and compound wheels C and D rotate independently about axis O. The wheels E and F rotate on pins fixed to the arm G. E gears with A and C and F gears with B and D as shown. All the wheels have the same module and the number of teeth are : $T_C = 28$; $T_D = 26$; $T_E = T_F = 18$. (CO5)
- (i) Find the number of teeth on A and B.
- (ii) If the arm G makes 100 r.p.m. clockwise and A is fixed, find the speed of B.



5. (a) Consider a disc cam with roller follower (no offset). Draw a suitable sketch and define the following terms for a cam :

- (i) base circle
- (ii) trace point
- (iii) pitch curve
- (iv) pressure angle
- (v) pitch point.

Clearly show the above on the sketch drawn.

(CO6)

(b) A cam with a minimum radius of 25 mm is to be designed for a knife-edge follower with the following data : (CO6)

- To raise the follower through 35 mm during 60° rotation of the cam.
- Dwell for the next 40° of the cam rotation.
- Descending of the follower during the next 90 degree of the cam rotation.
- Dwell during the rest of the cam rotation.

Draw the profile of the cam if the ascending and descending of the cam is with Simple Harmonic Motion and the line of stroke of the follower is offset 10 mm from the axis of the camshaft.

(c) Design a cam for operating the exhaust valve of an oil engine. It is required to give equal uniform acceleration and retardation during opening and closing of the valve each of which corresponds to 60° of cam rotation. The valve must remain in the fully open position for 20° of cam rotation. The lift of the valve is 37.5 mm and the least radius of the cam is 40 mm. The follower is provided with a roller of radius 20 mm and its line of stroke passes through the axis of the cam. (CO6)

Roll No.

TME-406

**B. TECH. (FOURTH SEMESTER)
END SEMESTER EXAMINATION, June/July, 2022**

MECHANICS OF MATERIALS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Deduce expression to determine the elongation of a trapezoidal section of uniform thickness. (CO1, CO2)

(b) What do you mean by thermal stresses ? Explain. (CO1, CO2)

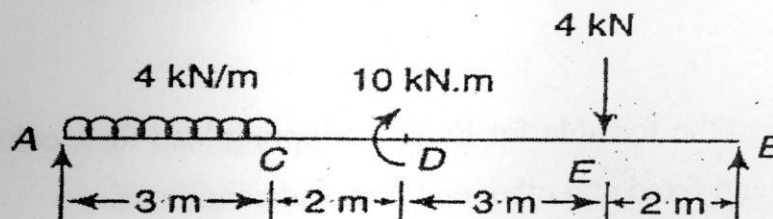
(c) Plot a tensile test diagram for steel. Explain its salient features.

(CO1, CO2)

2. (a) Derive the formula for Euler's crippling load for a column subjected to one end fixed and other end free, end conditions. (CO3)

P. T. O.

- (b) What assumptions are taken in the analysis of thin cylinders ? Deduce the expression for circumferential and longitudinal stresses developed in them. (CO3)
- (c) A 6m long thin cylindrical shell is 800 mm in diameter and 10 mm thick. It is subjected to an internal pressure of 4 MPa. Determine the change in diameter, change in length and change in volume of the shell. Take $E = 205 \text{ GPa}$ and Poisson's ratio $= 0.3$. (CO3)
3. (a) Deduce the torsion equation stating the assumptions made. (CO4)
- (b) What diameter of shaft will be required to transmit 80 kW at 80 rpm, if maximum torque is 30% greater than the mean and the limit of torsional stress is to be 56 MPa ? If modulus of rigidity is 80 GPa, what is the maximum angle of twist in 3 m length ? (CO4)
- (c) A hollow circular bar as a beam has outside diameter twice of the inside diameter. If it is subjected to a maximum bending moment of 40 kNm and the allowable bending stress is 100 MPa, determine the inside diameter of the bar. (CO4)
4. (a) Derive the expression for slope and deflection at the free end of a cantilever beam subjected to uniformly distributed load over the whole span of the beam. (CO5)
- (b) Draw shear force and bending moment diagram for a given beam. Also find the maximum value of bending moment. (CO5)



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- (c) A simply supported beam has a span of 15 m and carries two point loads of 4 kN and 9 kN at 6 m and 10 m respectively from one end. Find the deflection under each load and maximum deflection. (CO5)

Take $E = 200 \text{ GPa}$ and $\text{MOI (I)} = 400 \times 10^6 \text{ mm}^4$.

5. (a) What do you mean by principal planes and principal stresses ? Derive the expression for principal stresses for a body subjected to direct and shear stresses. (CO5, CO6)
- (b) Draw Mohr's stress circle indicating the radius and centre point coordinate for the given stress conditions : (CO5, CO6)
- (i) Pure shear (50 MPa).
 - (ii) Hydrostatic stresses (100 MPa).
 - (iii) Normal stress in X-direction 100 MPa (tensile) and Normal stress in Y- direction 50 MPa (tensile).
 - (iv) Normal stress in X-direction 100 MPa (tensile) and Normal stress in Y direction 50 MPa (compressive).
 - (v) Normal stress in X-direction 100 MPa (tensile) and Normal stress in Y direction 50 MPa (tensile) with 50 MPa shear stress.
- (c) A cantilever beam of 10 m span carries loads of 4 kN and 6 kN at 2 m and 6 m respectively from the fixed end along with another load of 6 kN at free end. Draw the shear force and bending moment diagram. (CO5, CO6)

Roll No.

TME-407

B. TECH. (ME) (FOURTH SEMESTER)

END SEMESTER EXAMINATION, June/July, 2022

FLUID MECHANICS AND FLUID MACHINERY

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Define and distinguish between the following set of fluid properties :

(CO1, CO2, CO3, CO4, CO5)

(i) Specific weight and mass density

(ii) Cohesion and adhesion

(iii) Dynamic viscosity and kinematic viscosity

P. T. O.

- (b) Define capillary effect. Derive the expression for rise of height in a capillary tube. (CO1, CO2, CO3, CO4, CO5)
- (c) A steel shaft of 25 mm diameter and 30 cm long falls of its own weight inside a vertical open tube of 25.2 mm diameter. The clearance assumed uniform is filled with oil of viscosity 1.5 Pa s. Calculate how fast the cylinder will fall at terminal conditions. Take density of steel as 7850 kg/m³. (CO1, CO2, CO3, CO4, CO5)
2. (a) State and prove the Bernoulli's equation from the principle of energy conservation. (CO1, CO2, CO4, CO5, CO3)
- (b) A 2 m long pipeline tapers uniformly from 10 cm diameter to 20 cm diameter at its upper end. The pipe center line slopes upwards at an angle of 30° to the horizontal and the flow direction is from smaller to bigger cross-section. If the pressure gauges installed at the lower and upper ends of the pipeline read 200 kPa and 230 kPa respectively, determine the flow rate and the fluid pressure at the mid-length of the pipeline. Assume no energy loss. (CO1, CO2, CO4, CO5, CO3)
- (c) Derive the expression for Discharge through an Orifice meter. (CO1, CO2, CO4, CO5, CO3)

3. (a) Define fundamental quantities, derived quantities. Explain the Principle of dimensional homogeneity. In terms of M-L-T determine the dimensions of radians, angular velocity, torque, work and power.

(CO1, CO2, CO3, CO4, CO6)

- (b) The resistance R due to wind on a tall vertical chimney is dependent upon the density ρ , viscosity μ , the wind velocity V , the diameter D , and height H of the chimney. Using Buckingham Pi theorem show that :

(CO1, CO2, CO3, CO4, CO6)

$$R = \rho V^2 D^2 f(\rho V D / \mu, D/H)$$

- (c) A 1 : 50 scale model of proposed dam is used to predict prototype flow conditions. If the design flood discharge over the spillway is 12000 m³/s, what water flow rate should be established in the model to simulate the flow ? Also if the velocity of 1.25 m/s is measured at a point in the model, what is the velocity at a corresponding point in the prototype ?

(CO1, CO2, CO3, CO4, CO6)

4. (a) What are the different flow regimes found in the flow of a real fluid ? Derive the expression for laminar flow in a circular pipe.

(CO1, CO2, CO3, CO4, CO5, CO6)

- (b) Define the major losses and minor losses during flow. Derive the expression losses due to sudden expansion and sudden contraction.

(CO1, CO2, CO3, CO4, CO5, CO6)

- (c) What do you understand by boundary layer ? Illustrate with reference to flow over a flat plate. (CO1, CO2, CO3, CO4, CO5, CO6)

5. (a) Define Impulse momentum principle and derive the expression for force with which a jet strikes a curved moving plate at centre.

(CO1, CO2, CO3, CO4, CO5, CO6)

- (b) Define Hydraulic Turbines. With the help of a neat sketch describe the construction and working of a Pelton wheel. Also derive the expression for the maximum efficiency. (CO1, CO2, CO3, CO4, CO5, CO6)

- (c) Define Hydraulic Pumps. With the help of a neat sketch explain the construction and working of a centrifugal pump and derive the expression for work done. (CO1, CO2, CO3, CO4, CO5, CO6)

Roll No.

TME-408

B. TECH. (ME) (FOURTH SEMESTER)

END SEMESTER EXAMINATION, June/July, 2022

APPLIED THERMODYNAMICS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

(v) Answer must be accompanied by neat and labelled diagrams.

(vi) Suitable assumptions can be made for any missing data.

(vii) Use of steam tables is permitted.

P. T. O.

1. (a) Describe the air standard petrol cycle and derive an expression for the efficiency of the same. (CO2)
- (b) Compare Otto, Diesel and Dual cycles on any three parameters. (CO3)
- (c) An air standard Carnot cycle is executed in a closed system between the temperature limits of 500 K and 1200 K. The pressure before and after the isothermal compression are 150 kPa and 350 kPa respectively. If the net work output per cycle is 0.6 kJ, determine (i) the maximum pressure in the cycle, (ii) the heat transfer to air. (CO4)
2. (a) Explain the concept of reheating and regeneration in gas turbine power plants with the help of diagrams. (CO2)
- (b) Derive expressions for the isentropic efficiencies of compressor and turbines with respect to the Brayton cycle. (CO3)
- (c) Air enters the compressor of an ideal air standard Brayton cycle at 100 kPa and 290 K with a mass flow rate of 6 kg/s. The compressor pressure ratio is 10. The turbine inlet temperature is 1500 K. If a regenerator with an effectiveness of 70% is incorporated in the cycle, evaluate the improvement in the thermal efficiency of the cycle. Take $\gamma = 1.5$ and constant pressure specific heat = 1.005 kJ/kg K. (CO4, CO6)
3. (a) Describe the concept of regeneration in Rankine cycle and discuss its overall impact on the cycle. (CO2, CO6)
- (b) Define and compare open and closed feed water heaters. (CO1, CO4)
- (c) A simple ideal Rankine cycle with water as the working fluid operates between the pressure limits of 5 MPa in the boiler and 20 kPa in the

(3)

condenser and a turbine inlet temperature of 500°C . The boiler is sized to provide a steam flow of 30 kg/s . Determine the power produced by the turbine and consumed by the pump. Use steam tables. (CO3, CO4)

4. (a) Explain multistage compressors and perfect intercooling. (CO5)
- (b) Derive an expression for the optimum pressure ratio in multistage compression. (CO5)
- (c) Calculate the diameter and stroke for a double acting single stage reciprocating air compressor of 50 kW having induction pressure 100 kN/m^2 and temperature 25°C . The law of compression is $PV^{1.3} = C$ and delivery pressure is 800 kN/m^2 . The revolution/sec = 15 and mean piston speed in 180 m/min . Clearance is neglected. (CO5)
5. (a) Define 1 ton of refrigeration. Describe the vapour compression refrigeration with the help of a neat diagram. (CO1, CO2)
- (b) Explain the vapour absorption refrigeration system with the help of a neat diagram. (CO3)
- (c) Sketch a typical psychrometric chart and define (i) Dry bulb temperature (ii) Wet bulb temperature (iii) Dew point temperature. (CO1, CO2)

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